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Camera device

Abstract

The present embodiment relates to a camera device including a magnet and a coil. The magnet includes: a first magnet disposed on each of first and second side surfaces of a camera module, and having a different polarity in a top portion and a bottom portion of the surface facing the coil; and a second magnet disposed on each of third and fourth side surfaces of the camera module, and having a different polarity on both side portions of the surface facing the coil. The coil includes: a first coil facing the first magnet; a second coil facing the first magnet and electrically isolated from the first coil; and a third coil facing the second magnet and electrically isolated from the first and second coils.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/593,901, filed Sep. 27, 2021; which is the U.S. national stage application of International Patent Application No. PCT/KR2020/003645, filed Mar. 17, 2020, which claims the benefit under 35 U.S.C. § 119 of Korean Application No. 10-2019-0035661, filed Mar. 28, 2019, the disclosures of each of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

(1) The present embodiment relates to a camera device.

BACKGROUND ART

(2) As various portable terminals are widely spread and commonly used, and the wireless Internet services have been commercialized, the demands of consumer related to portable terminals have been diversified and various kinds of additional devices have been installed in portable terminals.

(3) Among them, there is a camera device for photographing a subject as a photograph or a moving picture. Meanwhile, a camera device in recent years has been applied with a hand-shake correction function that inhibits an image from being shaken due to hand-shake of a photographer.

(4) However, the x-axis/y-axis direction lens shift used in the conventional hand-shake correction module has limitations in correcting various kinds of shaking.

DETAILED DESCRIPTION OF THE INVENTION

Technical Subject

(5) The present embodiment is intended to provide a camera device capable of correcting hand-shake for x-axis direction shift, y-axis direction shift, and rotation about the z-axis.

(6) In addition, it is intended to provide a camera device in which hand shake correction through a lens and hand shake correction through an image sensor are performed together.

Technical Solution

(7) The camera device according to the present embodiment comprises: a holder; a substrate disposed on the holder; a coil disposed on the substrate; a base disposed to be spaced apart from the holder; a magnet disposed on the base and facing the coil; and an image sensor coupled to the base, wherein the substrate comprises first to fourth corners, wherein the coil comprises a first coil disposed at the first corner of the substrate, a second coil disposed at the second corner of the substrate, and a third coil disposed at a third corner, and a fourth coil disposed at the fourth corner of the substrate, wherein a long side of the first coil and a long side of the third coil are disposed parallel to each other, a long side of the second coil and a long side of the fourth coil are disposed parallel to each other, and a long side of the first coil and a long side of the second coil may be disposed not to be parallel to each other.

(8) The long side of the first coil and the long side of the second coil may be disposed such that virtual extension lines thereof are orthogonal to each other.

(9) Currents may be independently applied to at least three among the first to fourth coils.

(10) The first to fourth coils may be electrically isolated from one another.

(11) The camera device may further comprise: a terminal portion comprising a terminal electrically connected to a terminal of the image sensor and disposed on the base; and a plurality of wires connecting the substrate and the terminal portion.

(12) The camera device may further comprise an image sensor substrate on which the image sensor is disposed, wherein the image sensor substrate is coupled to the terminal portion and the terminal portion is coupled to the base so that the image sensor can be coupled to the base.

(13) The plurality of wires may comprise a number of wires corresponding to the number of terminals of the image sensor.

(14) The plurality of wires may comprise a total of 24 wires, 6 between adjacent corners among the four corners of the base.

(15) The magnet may comprise: a first magnet facing the first coil and disposed at a first corner of the base; a second magnet facing the second coil and disposed at a second corner of the base; a third magnet facing the third coil and disposed at a third corner of the base; and a fourth magnet facing the fourth coil and disposed at a fourth corner of the base.

(16) The base comprises a first side surface and a second side surface disposed opposite to each other, and a third side surface and a fourth side surface disposed opposite to each other between the first side surface and the second side surface, wherein the first corner of the base is disposed between the first side surface and the third side surface, the second corner of the base is disposed between the third side surface and the second side surface, the third side surface of the base a corner is disposed between the second side surface and the fourth side surface, and the fourth corner of the base is disposed between the fourth side surface and the first side surface, and wherein the polarity of the surface facing the coil of the first magnet is different from a portion close to the first side surface and a portion close to the second side surface, and the polarity of a surface of the second magnet facing the coil may be different between a portion close to the third side surface and a portion close to the fourth side surface.

(17) Any one of a forward current and a reverse current may be selectively applied to each of the four coils.

(18) The camera device according to the present embodiment comprises an optical module for performing a hand-shake correction function; an image sensor disposed in alignment with the optical module; and an actuator for driving the image sensor, wherein the actuator can move the image sensor in a first direction perpendicular to the optical axis, move in a second direction perpendicular to the optical axis and the first direction, and rotate based on the optical axis.

(19) The optical module may comprise one or more of a liquid lens and a MEMS actuator.

(20) A camera device according to the present embodiment comprises: a base assembly comprising an image sensor; a holder assembly comprising a lens and spaced apart from the base assembly; a

coil disposed on the holder assembly; a magnet disposed on the base assembly and facing the coil; and a plurality of elastic members connecting the base assembly and the holder assembly, wherein the plurality of elastic members may be electrically connected to the image sensor.

Advantageous Effects

(21) Through the present embodiment, an x-axis direction shift, a y-axis direction shift, and a rotation about the z-axis corresponding to hand-shake may be performed with respect to the image sensor.

(22) In addition, hand-shake correction for the image sensor and hand-shake correction for a corresponding lens can be performed together.

(23) Through this, it is possible to provide a more enhanced hand-shake correction function.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a perspective view of a camera device according to the present embodiment.

(2) FIG. 2 is a cross-sectional view viewed from A-A of FIG. 1.

(3) FIG. 3 is a cross-sectional view viewed from B-B of FIG. 1.

(4) FIG. 4 is an exploded perspective view of a partial configuration of the camera device according to the present embodiment.

(5) FIG. 5 is an exploded perspective view of a partial configuration of the camera device according to the present embodiment.

(6) FIG. 6 is a bottom perspective view of a partial configuration of the camera device according to the present embodiment.

(7) FIG. 7 is a perspective view of a partial configuration of the camera device according to the present embodiment.

(8) FIG. 8 is a perspective view of a partial configuration of a camera device according to a modified embodiment.

(9) FIG. 9a is an exploded perspective view of a partial configuration of the camera device according to the present embodiment.

(10) FIG. 9b is a cross-sectional view of a state in which a partial configuration of the camera device of FIG. 9a is combined when viewed from C-C.

(11) FIG. 9c is a cross-sectional view of a state in which a partial configuration of the camera device of FIG. 9a is combined when viewed from D-D.

(12) FIG. 10 and FIG. 11 are exploded perspective views of a partial configuration of the camera device according to the present embodiment when viewed from a different direction than that of FIG. 9a.

(13) FIG. 12 is an exploded perspective view of an image sensor module of the camera device according to the present embodiment.

(14) FIG. 13 is an exploded perspective view of an image sensor module of the camera device according to the present embodiment when viewed from a different direction than that of FIG. 12.

(15) FIG. 14 is a diagram for explaining the x-axis direction shift driving through a partial configuration of a camera device according to the present embodiment.

(16) FIG. 15 is a diagram for explaining y-axis direction shift driving through a partial configuration of a camera device according to the present embodiment.

(17) FIG. 16 is a view for explaining z-axis rotational driving through a partial configuration of a camera device according to the present embodiment.

(18) FIG. 17 (a) is a view illustrating a magnet disposed on the base together with the x-axis and y-axis, and FIG. 17 (b) is a view illustrating a base, a magnet and a coil with the z-axis rotational driving.

(19) FIG. 18 is a view illustrating a magnetic flow and a Lorentz Force between a magnet and a coil of a camera device according to the present embodiment.

(20) FIG. 19 is a perspective view of an optical device according to the present embodiment.

(21) FIG. 20 is a block diagram of an optical device illustrated in FIG. 19.

BEST MODE

(22) Hereinafter, preferred embodiments of the present invention will be described in detail with reference to the accompanying drawings.

(23) However, the technical idea of the present invention is not limited to some embodiments to be described, but may be implemented in various forms, and within the scope of the technical idea of the present invention, one or more of the constituent elements may be selectively combined or substituted between embodiments.

(24) In addition, the terms (comprising technical and scientific terms) used in the embodiments of the present invention, unless explicitly defined and described, can be interpreted as a meaning that can be generally understood by a person skilled in the art, and commonly used terms such as terms defined in the dictionary may be interpreted in consideration of the meaning of the context of the related technology.

(25) In addition, terms used in the present specification are for describing embodiments and are not intended to limit the present invention.

(26) In the present specification, the singular form may comprise the plural form unless specifically stated in the phrase, and when described as “at least one (or more than one) of A and B and C”, it may comprise one or more of all combinations that can be combined with A, B, and C

(27) In addition, in describing the components of the embodiment of the present invention, terms such as first, second, A, B, (a), and (b) may be used. These terms are merely intended to distinguish the components from other components, and the terms do not limit the nature, order or sequence of the components.

(28) And, when a component is described as being ‘connected’, ‘coupled’ or ‘interconnected’ to another component, the component is not only directly connected, coupled or interconnected to the other component, but may also comprise cases of being ‘connected’, ‘coupled’, or ‘interconnected’ due that another component between that other components.

(29) In addition, when described as being formed or arranged in “on (above)” or “below (under)” of each component, “on (above)” or “below (under)” means that it comprises not only the case where the two components are directly in contact with, but also the case where one or more other components are formed or arranged between the two components. In addition, when expressed as “on (above)” or “below (under)”, the meaning of not only an upward direction but also a downward direction based on one component may be comprised.

(30) An ‘optical axis (refer to OA of FIG. 4) direction’ used hereinafter is defined as an optical axis direction of a lens and/or an image sensor coupled to the lens driving device.

(31) The ‘vertical direction’ used hereinafter may be a direction parallel to the optical axis direction. The vertical direction may correspond to the ‘z-axis direction (refer to FIG. 4)’. The ‘horizontal direction’ used below may be a direction perpendicular to the vertical direction. That is, the horizontal direction may be a direction perpendicular to the optical axis. Accordingly, the horizontal direction may comprise an ‘x-axis direction’ and a ‘y-axis direction’ (refer to FIG. 4).

(32) The ‘autofocus function’ used hereinafter is defined as a function of automatically focusing on the subject by adjusting the distance from the image sensor by moving the lens in the optical axis direction according to the distance of the subject so that a clear image of the subject can be obtained on the image sensor. Meanwhile, ‘auto focus’ may correspond to ‘AF (Auto Focus)’.

(33) A ‘hand-shake correction function’ used hereinafter is defined as a function of moving a lens and/or an image sensor to offset vibration (movement) generated in the image sensor by an external force. Meanwhile, ‘hand-shake correction’ may correspond to ‘optical image stabilization (OIS)’.

(34) Hereinafter, the configuration of the camera device will be described with reference to the

drawings.

(35) FIG. 1 is a perspective view of a camera device according to the present embodiment, FIG. 2 is a cross-sectional view viewed from A-A of FIG. 1, FIG. 3 is a cross-sectional view viewed from B-B of FIG. 1, FIG. 4 is an exploded perspective view of a partial configuration of the camera device according to the present embodiment, FIG. 5 is an exploded perspective view of a partial configuration of the camera device according to the present embodiment, FIG. 6 is a bottom perspective view of a partial configuration of the camera device according to the present embodiment, FIG. 7 is a perspective view of a partial configuration of the camera device according to the present embodiment, FIG. 8 is a perspective view of a partial configuration of a camera device according to a modified embodiment, FIG. 9a is an exploded perspective view of a partial configuration of the camera device according to the present embodiment, FIG. 9b is a cross-sectional view of a state in which a partial configuration of the camera device of FIG. 9a is combined when viewed from C-C, FIG. 9c is a cross-sectional view of a state in which a partial configuration of the camera device of FIG. 9a is combined when viewed from D-D, FIG. 10 and FIG. 11 are exploded perspective views of a partial configuration of the camera device according to the present embodiment when viewed from a different direction than that of FIG. 9a, FIG. 12 is an exploded perspective view of an image sensor module of the camera device according to the present embodiment, FIG. 13 is an exploded perspective view of an image sensor module of the camera device according to the present embodiment when viewed from a different direction than that of FIG. 12, FIG. 14 is a diagram for explaining the x-axis direction shift driving through a partial configuration of a camera device according to the present embodiment, FIG. 15 is a diagram for explaining y-axis direction shift driving through a partial configuration of a camera device according to the present embodiment, FIG. 16 is a view for explaining z-axis rotational driving through a partial configuration of a camera device according to the present embodiment, FIG. 17 (a) is a view illustrating a magnet disposed on the base together with the x-axis and y-axis, and FIG. 17 (b) is a view illustrating a base, a magnet and a coil with the z-axis rotational driving, and FIG. 18 is a view illustrating a magnetic flow and a Lorentz Force between a magnet and a coil of a camera device according to the present embodiment.

(36) The camera device 10A may comprise a camera module. The camera device 10A may comprise a lens driving device. The lens driving device may be a voice coil motor (VCM). The lens driving device may be a lens driving motor. The lens driving device may be a lens driving actuator. The lens driving device may comprise an AF module. The lens driving device may comprise an OIS module.

(37) The camera device 10A may comprise an actuator. The actuator may drive the image sensor 444. The actuator may tilt the image sensor 444. The actuator may move the image sensor 444. The actuator may rotate the image sensor 444. The actuator may move the image sensor 444 in a first direction perpendicular to the optical axis, move the image sensor 444 in a second direction perpendicular to the optical axis and the first direction, and rotate the image sensor 444 based on the optical axis. At this time, the first direction may be the x-axis direction, the second direction may be the y-axis direction, and the optical axis may be the z-axis direction. The actuator may comprise a coil 310 and a magnet 320. The actuator may move the image sensor 444 through electromagnetic force.

(38) The camera device 10A may comprise a holder 110. The holder 110 may be disposed on a lower surface of a printed circuit board 50. The holder 110 may comprise a protrusion for fitting into the groove of the printed circuit board 50. The holder 110 may be disposed on an upper surface of the substrate 120. The holder 110 may be disposed between the printed circuit board 50 and the substrate 120. A lens module 210 may be disposed in the holder 110. An optical module may be disposed on the holder 110. The holder 110 may be coupled to the housing 600.

(39) The holder 110 may comprise a step 111. The step 111 may be formed in the circumference of the hole 112 of the holder 110. A lens module 210 may be disposed on the step 111. The step 111

may support a lower surface of a portion of the lens module **210**. Through this, it is possible to inhibit the lens module **210** from being displaced downward while seated on the step **111**.

(40) The holder **110** may comprise a hole **112**. The hole **112** may be a hollow hole. The hole **112** may be an opening. A lens module **210** may be disposed in the hole **112**. A portion of the lens module **210** may be extended below the step **111** through the hole **112**.

(41) The holder **110** may comprise a first hole **113**. The first hole **113** may be formed to avoid a portion of the substrate **120** to be coupled with the wire **510**. The first hole **113** may comprise a plurality of holes. The first hole **113** may comprise two holes. The holder **110** may comprise a second hole **114**. The second hole **114** may be formed to avoid the sensor **520** being coupled to the substrate **120**. The second hole **114** may comprise a plurality of holes. The second hole **114** may comprise four holes.

(42) The holder **110** may comprise a first groove **115**. The first groove **115** may be formed to avoid a portion of the substrate **120** to be coupled to the wire **510**. The first groove **115** may be formed on a side surface of the holder **110**. The first groove **115** may be formed on each of both side surfaces of the holder **110**. The first groove **115** may comprise a plurality of grooves. The first groove **115** may comprise two grooves. The holder **110** may comprise a second groove **116**. The second groove **116** may be formed in a shape corresponding to the protrusion of the housing **600** so that it can be matched with the protrusion of the housing **600**. However, the second groove **116** may not be provided in a shape corresponding to the protrusion of the housing **600**. The second groove **116** may be formed on a side surface of the holder **110**. The second groove **116** may be formed on each of both sides of the holder **110**. The second groove **116** may comprise a plurality of grooves. The second groove **116** may comprise three grooves. The second groove **116** is formed as two grooves on one side surface of the holder **110** and may be formed as a single groove connected with the two grooves on the other side surface of the holder **110**.

(43) The camera device **10A** may comprise a substrate **120**. The substrate **120** may be disposed in the holder **110**. The substrate **120** may be disposed on a lower surface of the holder **110**. The upper surface of the substrate **120** may be in contact with a lower surface of the holder **110**. The substrate **120** may be disposed below the printed circuit board **50**. The substrate **120** may be coupled to the wire **510**. The substrate **120** may be a rigid flexible PCB (RFPCB). The substrate **120** may comprise first to fourth corners.

(44) The substrate **120** may comprise a first hole **121**. The first hole **121** may be formed in the central portion of the substrate **120**. The first hole **121** may be a hollow hole. The first hole **121** may be an opening. A lens module **210** may be disposed in the first hole **121**. The first hole **121** of the substrate **120** may be formed to have a greater width than the hole **112** of the holder **110**.

(45) The substrate **120** may comprise a coupling portion **122**. The substrate **120** may be coupled to the wire **510** at the coupling portion **122**. The substrate **120** and the wire **510** may be coupled through soldering. The coupling portion **122** may be a portion in which a solder resistor is opened to be electrically connected to the wire **510**. A second hole **123** may be formed in the coupling portion **122**. The substrate **120** may comprise a second hole **123**. The second hole **123** may be a wire through hole through which the wire **510** passes.

(46) The substrate **120** may comprise a connector **124**. The connector **124** may be electrically connected to the printed circuit board **50**. A connector corresponding to the connector **124** of the substrate **120** may be disposed on the printed circuit board **50**. The connector **124** may comprise a port for electrically connecting to an external device.

(47) The substrate **120** may comprise a terminal **125**. The terminal **125** may be formed on a lower surface of the substrate **120**. The terminal **125** may be electrically connected to the coil **310**. The terminal **125** may be coupled to a pair of lead wires of the coil **310** by soldering or Ag epoxy. The terminal **125** may comprise a plurality of terminals. The terminal **125** may comprise a total of eight terminals, two for each of four coils.

(48) The camera device **10A** may comprise a lens module **210**. The lens module **210** may be

disposed in the holder **110**. The lens may be disposed at a position corresponding to the image sensor **444**. The lens module **210** may comprise at least one lens. The lens module **210** may comprise a plurality of lenses. The lens module **210** may comprise five lenses. The lens module **210** may comprise first to fifth lenses **211**, **212**, **213**, **214**, and **215**. The lens module **210** may comprise a barrel **216**. A plurality of lenses may be disposed inside the barrel **216**. The lens module **210** may comprise a hole **217**. An optical module may be disposed in the hole **217** of the lens module **210**. The hole **217** of the lens module **210** may be formed to penetrate through the lens module **210** in the horizontal direction between the plurality of lenses. Through this, the optical axis of the plurality of lenses and the optical axis of the optical module may be aligned. The hole **217** of the lens module **210** may be formed between the second lens **212** and the third lens **213**.

(49) The camera device **10A** may comprise an optical module. The optical module may perform hand-shake correction (OIS) function. The optical module may perform an auto focus (AF) function. The optical module may be disposed in alignment with the plurality of lenses and the image sensor **444**. The optical module may be disposed between the plurality of lenses. The optical module may be disposed between the second lens **212** and the third lens **213**.

(50) The camera module may comprise a variable lens. The variable lens may be a variable focus lens. The variable lens may be a lens whose focus is controlled. The focus may be adjusted by moving the lens and/or changing the shape of the lens. The variable lens may comprise at least one among a liquid lens **687**, a polymer lens, a liquid crystal lens, a voice coil motor (VCM) actuator, a shape memory alloy (SMA) actuator, and a micro electro mechanical systems (MEMS) actuator. The liquid lens **687** may comprise at least one of a liquid lens **687** containing one type of liquid and a liquid lens **687** containing two types of liquid. The liquid lens **687** comprising one type of liquid may change the focus by adjusting a membrane disposed at a position corresponding to the liquid. For example, the focus can be changed by pressing the membrane by the electromagnetic force of the magnet and coil. The liquid lens **687** comprising two types of liquids may comprise a conductive liquid and a non-conductive liquid. In this case, the focus may be changed by adjusting the interface formed between the conductive liquid and the non-conductive liquid using a voltage applied to the liquid lens **687**. The polymer lens can change the focus by controlling a polymer material through a driving unit such as a piezo. The liquid crystal lens can change the focus by controlling the liquid crystal by electromagnetic force. The VCM actuator can change focus by moving a solid lens or a lens assembly comprising a solid lens through electromagnetic force between a magnet and a coil. The SMA actuator may change a focus by moving a solid lens or a lens assembly comprising a solid lens using a shape memory alloy. The MEMS actuator **220** may change the focus by moving a solid lens or a lens assembly comprising the solid lens through electrostatic force generated when voltage is applied. The optical module may comprise one or more of the liquid lens **220a** and the MEMS actuator **220**.

(51) The barrel **216** may comprise a hole **217** penetrating through the barrel **216** in a horizontal direction. At this time, the variable lens may be disposed by being inserted into the hole **217** formed in the barrel **216**. Meanwhile, the variable lens may be electrically connected to the printed circuit board **50**. The variable lens may be electrically connected to the printed circuit board **50** through the substrate.

(52) As illustrated in FIG. 7, the camera device **10A** may comprise a MEMS actuator **220**. The MEMS actuator **220** may perform an autofocus function and/or an hand-shake correction function by moving a moving lens using a silicon wafer.

(53) The MEMS actuator **220** may be connected to the substrate **221**. The substrate **221** may comprise a terminal **222**. The terminal **222** may comprise a plurality of terminals. Terminal **222** may comprise six terminals. The terminal **222** of the substrate **221** may be connected to the terminal **50a** of the printed circuit board **50**.

(54) As a modified embodiment, as illustrated in FIG. 8, the camera device **10A** may comprise a liquid lens **220a**. The liquid lens **220a** may be disposed on the lens module **210**. The liquid lens

220a may be connected to the first substrate **221** and the second substrate **223**. Each of the first substrate **221** and the second substrate **223** may comprise a terminal **222**. Four terminals **222** may be provided on a side of the first substrate **221** and one terminal **222** may be provided on a side of the second substrate **223**. The four terminals **222** of the first substrate **221** are electrodes electrically connected to the four individual electrodes of the liquid lens **220a**, and one terminal **222** of the second substrate **223** may be an electrode electrically connected to one common electrode of the liquid lens **220a**.

(55) The liquid lens **220a** whose focal length is adjusted in response to the driving voltage may receive an operating voltage through an upper terminal. The upper terminal may have the same angular distance and may comprise four individual terminals disposed in different directions. When an operating voltage is applied through the upper terminal, the interface between the conductive liquid and the non-conductive liquid formed in the lens region may be deformed. The lower terminal may be a common terminal. The upper terminal may be an upper electrode. The lower terminal may be a lower electrode. The liquid lens **220a** may be spaced apart from the solid lens. In this embodiment, epoxy may be applied through the space between the liquid lens **220a** and the solid lens, and active alignment of the liquid lens **220a** may be performed. Active alignment of the liquid lens **220a** and the solid lens can be performed by applying a current to the liquid lens **220a** in a state in which the epoxy is applied, aligning the optical axis with the solid lens, pre-curing the epoxy, and then main-curing the epoxy.

(56) The liquid lens **220a** may be described as a plurality of capacitors having one side applied with an operating voltage from an upper terminal and the other side connected to a lower terminal. Here, the plurality of capacitors comprised in the equivalent circuit may have a small capacitance of about 200 pF. In the present embodiment, the upper terminal of the liquid lens **220a** is an individual terminal and the lower terminal may be a common terminal. Alternatively, the upper terminal of the liquid lens **220a** may be a common terminal and the lower terminal may be an individual terminal. In the liquid lens **220a**, the interface formed between the conductive liquid and the non-conductive liquid may be deformed by the current applied to the upper terminal and the lower terminal. Through this, any one or more of the AF function and the OIS function can be performed.

(57) The camera device **10A** may comprise a coil **310**. The coil **310** may be disposed on the substrate **120**. The coil **310** may be electrically connected to the substrate **120**. The coil **310** may be disposed to face the magnet **320**. When a current is applied to the coil **310**, an electric field may be formed around the coil **310**. When a current is applied to the coil **310**, any one of the coil **310** and the magnet **320** may move relative to the other through electromagnetic interaction between the coil **310** and the magnet **320**.

(58) The coil **310** may comprise four coils. Currents may be independently applied to at least three coils among the four coils. In the first embodiment, the coil **310** may be controlled by three channels. Alternatively, in the second embodiment, the coil **310** may be controlled by four channels. The four coils **310** may be electrically isolated from each other. Any one of a forward current and a reverse current may be selectively applied to each of the four coils **310**. In the present embodiment, only three of the four coils are electrically isolated, and one coil may be electrically connected to the other coil. Alternatively, all four coils may be electrically isolated. When only 3 out of 4 coils are electrically isolated, 3 pairs of 6 lead wires come out from the coil **310**, and when all four coils are electrically isolated, a total of eight lead wires in four pairs may come out from the coil **310**.

(59) When controlling 4 coils with 3 channels as in the first embodiment of the present embodiment, it should be driven with one pair of the coil **310** and the magnet **320** in rotational driving about z-axis, but when controlling 4 coils with 4 channels as in the second embodiment, it can be driven with two pairs of the coil **310** and the magnet **320** in rotational driving about z-axis.

(60) The coil **310** may comprise first to fourth coils **311**, **312**, **313**, and **314**. The first coil **311** may

be disposed to face the first magnet **321**. The second coil **312** may be disposed to face the second magnet **322**. The third coil **313** may be disposed to face the third magnet **323**. The fourth coil **314** may be disposed to face the fourth magnet **324**. The first coil **311** may be disposed at a first corner of the substrate **120**. The second coil **312** may be disposed at the second corner of the substrate **120**. The third coil **313** may be disposed at the third corner of the substrate **120**. The fourth coil **314** may be disposed at the fourth corner of the substrate **120**. The first coil **311** and the third coil **313** are disposed on a first diagonal direction of the substrate **120**, and the second coil **312** and the fourth coil **314** may be disposed on a second diagonal direction of the substrate **120**.

(61) In the present embodiment, the first coil **311** and the third coil **313** are disposed to be elongated in the first direction, and the second coil **312** and the fourth coil **314** may be disposed to be elongated in the second direction. At this time, the first direction and the second direction may be perpendicular. A long side of the first coil **311** and a long side of the third coil **313** may be disposed parallel to each other. The long side of the second coil **312** and the long side of the fourth coil **314** may be disposed parallel to each other. The long side of the first coil **311** and the long side of the second coil **312** may be disposed not to be parallel to each other. At this time, the long side of the first coil **311** and the long side of the second coil **312** may be disposed such that virtual extension lines thereof are perpendicular to each other. The arrangement direction of the first coil **311** and the arrangement direction of the second coil **312** may be orthogonal to each other. The long side of the coil may mean a long side at the outer circumference of the coil. The short side of the coil may mean a short side at the outer circumference of the coil. The outer circumference of the coil may comprise two long sides and two short sides. A corner portion where the long side and the short side meet may be formed to be round.

(62) The first coil **311** may be disposed in a direction different from that of the second coil **312**. The first coil **311** may be disposed in the same direction as the third coil **313**. The second coil **312** may be disposed in the same direction as the fourth coil **314**.

(63) In the present embodiment, current may be independently applied to at least three among the first to fourth coils **311**, **312**, **313**, and **314**. The first to fourth coils **311**, **312**, **313**, and **314** may be electrically isolated from one another.

(64) The camera device **10A** may comprise a magnet **320**. The magnet **320** may be disposed on the base **410**. The magnet **320** may be disposed at a corner of the base **410**. The magnets **320** may be respectively disposed at four corners of the base **410**. The magnet **320** may face the coil **310**. The magnet **320** may electromagnetically interact with the coil **310**. The magnet **320** may move through electromagnetic interaction with the coil **310**. That is, when a current is applied to the coil **310**, the magnet **320** may move. The magnet **320** may be a flat magnet having a flat plate shape. In the present embodiment, the coil **310** is fixed and the magnet **320** may move. However, as a modified embodiment, the coil **310** and the magnet **320** may be disposed in reverse.

(65) The magnet **320** may comprise a plurality of magnets. The magnet **320** may comprise four magnets. The magnet **320** may comprise first to fourth magnets **321**, **322**, **323**, and **324**. The first magnet **321** may face the first coil **311**. The first magnet **321** may be disposed at a first corner **410e** of the base **410**. The second magnet **322** may face the second coil **312**. The second magnet **322** may be disposed at a second corner **410f** of the base **410**. The third magnet **323** may face the third coil **313**. The third magnet **323** may be disposed at a third corner **410g** of the base **410**. The fourth magnet **324** may face the fourth coil **314**. The fourth magnet **324** may be disposed at a fourth corner **410h** of the base **410**. Each of the plurality of magnets may be disposed perpendicularly to an adjacent magnet and parallel to the magnets disposed in a diagonal direction.

(66) A polarity of a surface of the first magnet **321** facing the coil **310** may be different from a portion close to a first side surface **410a** and a portion close to a second side surface **410b**. A polarity of a surface of the second magnet **322** facing the coil **310** may be different from a portion close to a third side surface **410c** and a portion close to a fourth side portion **410d**. A polarity of a surface of the third magnet **323** facing the coil **310** may be different from a portion close to the first

side surface **410a** and a portion close to the second side surface **410b**. A polarity of a surface of the fourth magnet **324** facing the coil **310** may be different from a portion close to the third side surface **410c** and a portion close to the fourth side surface **410d**. That is, the first magnet **321** and the third magnet **323** may be disposed in the same direction, and the second magnet **322** and the fourth magnet **324** may be disposed in the same direction. The first magnet **321** may be vertically disposed with the second magnet **322**. Polarities of the first to fourth magnets **321**, **322**, **323**, and **324** may be the same as inner side portions. The polarities of the first to fourth magnets **321**, **322**, **323**, and **324** may be the same as the outer side portions. The polarity of each of the first to fourth magnets **321**, **322**, **323**, and **324** may be formed as an N pole at an inner side portion. The polarity of each of the first to fourth magnets **321**, **322**, **323**, and **324** may be formed as an S pole at an outer side portion thereof. However, in a modified embodiment, the polarity of each of the first to fourth magnets **321**, **322**, **323**, and **324** may have an inner side portion formed as an S pole and an outer side portion formed as an N pole.

(67) As illustrated in FIG. **14**, in the present embodiment, when a current in the same direction is applied to the second coil **312** and the fourth coil **314**, the image sensor **444** coupled to the base **410** may be moved (shifted) in the x-axis direction through electromagnetic interaction with the second magnet **322** and the fourth magnet **324**, respectively. That is, the second coil **312** and the second magnet **322** and the fourth coil **314** and the fourth magnet **324** may be used for shift driving of the image sensor **444** in the x-axis direction. At this time, the second coil **312** and the second magnet **322** is a first x-axis shift driving unit X2, and the fourth coil **314** and the fourth magnet **324** may be a second x-axis shift driving unit X1.

(68) As illustrated in FIG. **15**, in the present embodiment, when currents in the same direction are applied to the first coil **311** and the third coil **313**, the image sensor **444** coupled to the base **410** may be moved (shifted) in the y-axis direction through electromagnetic interaction between the first magnet **321** and the third magnet **323**, respectively. That is, the first coil **311** and the first magnet **321**, and the third coil **313** and the third magnet **323** may be used for shift driving of the image sensor **444** in the y-axis direction. In this case, the first coil **311** and the first magnet **321** may be a first y-axis shift driving unit Y1, and the third coil **313** and the third magnet **323** may be a second y-axis shift driving unit Y2.

(69) As illustrated in FIG. **16**, the present embodiment, when currents of opposite directions are applied to the first coil **311** and the third coil **313** and currents of opposite directions are applied to the second coil **312** and the fourth coil **314**, and at this time, if the direction in which the magnet **320** is rotated by the current applied to the first coil **311** and the current applied to the second coil **312** is the same, the image sensor **444** coupled to the base **410** may be rotating (rolling) about the z-axis. The embodiment illustrated in FIG. **16** illustrates a case in which the coil **310** is controlled by 4 channels, and if the coil **310** is controlled by 3 channels the image sensor **444** can be rolled through the first coil **311** and the third coil **313** or the second coil **312** and the fourth coil **314**. This is because if there is a coil bundled with one channel among the first coil **311** and the third coil **313**, and the second coil **312** and the fourth coil **314**, current cannot be applied in the opposite direction.

(70) As illustrated in FIG. **17 (b)**, in the present embodiment, a forward current is applied to the first coil **311**, and through this, the first coil **311** pushes the first magnet **321** in a first direction (refer to a in FIG. **17 (b)**), a forward current is applied to the second coil **312** and the second coil **312** pushes the second magnet **322** in a second direction (refer to b in FIG. **17 (b)**), a reverse current is applied to the third coil **313**, and through this, the third coil **313** pushes the third magnet **323** in a third direction (refer to c in FIG. **17 (b)**), a reverse current is applied to the fourth coil **314**, and through this, the fourth coil **314** pushes the fourth magnet **324** in a fourth direction (refer to d in FIG. **17 (b)**), and as a results, the image sensor **444** coupled to the base **410** may be rotated about the z-axis (refer to e of FIG. **17 (b)**). At this time, the first to fourth directions may correspond to a clockwise direction with respect to the center of the base **410**.

(71) In the present embodiment, the magnetic flow of the magnet **320** is same as illustrated in FIG.

18. Referring to FIG. 18, it can be confirmed that there is a magnetic force line passing perpendicular to the coil 310, and in this state, when a current is applied to the coil 310, the coil 310 can be moved against the magnet 320 according to the Lorentz Force.

(72) The camera device 10A may comprise a base 410. The base 410 may be disposed to be spaced apart from the holder 110. The base 410 is a part that moves together with the magnet 320 when a current is applied to the coil 310 and may be a mover. In addition, the base 410 may be a sensor PCB holder. The base 410 may be shifted in the x-axis direction. The base 410 may be shifted in the y-axis direction. The base 410 may be rotated about the z-axis (optical axis).

(73) The base 410 may comprise a first hole 411. The first hole 411 may be a hollow hole. The first hole 411 may be an opening.

(74) The base 410 may comprise a groove 412. The groove 412 may be formed on an upper surface of the base 410. The groove 412 may accommodate at least a portion of the magnet 320. The magnet 320 may be disposed in the groove 412 of the base 410. The groove 412 may be formed in a shape corresponding to the magnet 320. However, the depth of the groove 412 may be smaller than the thickness in the corresponding direction of the magnet 320. In this case, a portion of the magnet 320 disposed in the groove 412 may be protruded from the base 410. The groove 412 may comprise a plurality of grooves. The number of grooves 412 may be formed to correspond to the number of magnets 320. The groove 412 may comprise four grooves.

(75) The base 410 may comprise a second hole 413. The second hole 413 may be formed penetrating through the base 410 in a direction parallel to the optical axis. A wire 510 may be disposed in the second hole 413. The wire 510 may pass through the second hole 413. The second hole 413 may comprise a plurality of holes. The second hole 413 may be formed in a number corresponding to the number of wires 510. The second hole 413 may comprise 24 holes.

(76) The base 410 may comprise a first protrusion 414. The first protrusion 414 may be formed on a lower surface of the base 410. The first protrusion 414 may be inserted into a first hole 421 of a reinforcing member 420 and a hole 431-1 of a terminal portion 430. The first protrusion 414 may be formed in a shape corresponding to the first hole 421 of the reinforcing member 420 and the hole 431-1 of the terminal portion 430. The first protrusion 414 may comprise a plurality of protrusions. The first protrusion 414 may comprise four protrusions. The four protrusions may be respectively formed at the four corners of the base 410.

(77) The base 410 may comprise a second protrusion 415. The second protrusion 415 may be formed on a lower surface of the base 410. The second protrusion 415 may be spaced apart from the first protrusion 414. The second protrusion 415 may be extended from a side surface of the base 410. A lower surface of the second protrusion 415 may be disposed lower than a lower surface of the reinforcing plate 445 of the image sensor module 440. The second protrusion 415 may comprise a plurality of protrusions. The second protrusion 415 may comprise four protrusions. The four protrusions may be respectively formed at the four corners of the base 410.

(78) The base 410 may comprise a guide protrusion 416. The guide protrusion 416 may be formed on a lower surface of the base 410. The guide protrusion 416 may guide an assembly position of the image sensor module 440. The guide protrusion 416 may be in contact with the cover 441 of the image sensor module 440. The guide protrusion 416 may be in contact with four side surfaces of the cover 441 of the image sensor module 440.

(79) The base 410 may comprise a plurality of side surfaces. The base 410 may comprise four side surfaces. The base 410 may comprise first to fourth side surfaces 410a, 410b, 410c, and 410d. The base 410 may comprise a first side surface 410a and a second side surface 410b disposed opposite to each other, and a third side surface 410c and a fourth side surface 410d disposed opposite to each other between the first side surface 410a and a second side surface 410b.

(80) The base 410 may comprise corners formed between a plurality of side surfaces. The base 410 may comprise a plurality of corners. The base 410 may comprise four corners. The base 410 may comprise first to fourth corners 410e, 410f, 410g, and 410h. The first corner 410e of the base 410

may be disposed between the first side surface **410a** and the third side surface **410c**. The second corner **410f** of the base **410** may be disposed between the third side surface **410c** and the second side surface **410b**. The third corner **410g** of the base **410** may be disposed between the second side surface **410b** and the fourth side surface **410d**. The fourth corner **410h** of the base **410** may be disposed between the fourth side surface **410d** and the first side surface **410a**.

(81) The camera device **10A** may comprise a reinforcing member **420**. The reinforcing member **420** may be formed of a SUS. The reinforcing member **420** may reinforce the terminal portion **430**. The reinforcing member **420** may be coupled to the terminal portion **430**. The reinforcing member **420** may be bonded to the terminal portion **430** by an adhesive. The reinforcing member **420** may be disposed on a lower surface of the base **410**.

(82) The reinforcing member **420** may comprise a first hole **421**. The first hole **421** may be coupled to the first protrusion **414** of the base **410**. The reinforcing member **420** may comprise a second hole **422**. An adhesive may be applied to the second hole **422**. The second hole **422** may be formed in a protruding portion of the reinforcing member **420**. The second hole **422** may comprise a plurality of holes. The second hole **422** may be formed in a total of 16, two in each of the four corners of the reinforcing member **420**, two in eight protruding portions in total.

(83) The reinforcing member **420** may comprise a protrusion **423**. The protrusion **423** may be formed to be protruded inward from a corner of the reinforcing member **420**. A space in which the first hole **421** is to be formed may be secured through the protrusion **423**. A first hole **421** may be formed in the protrusion **423**.

(84) The camera device **10A** may comprise a terminal portion **430**. The terminal portion **430** may be disposed on a lower surface of the base **410**. The terminal portion **430** may be coupled to the reinforcing member **420**. The terminal portion **430** may be coupled to the image sensor module **440**.

(85) The terminal portion **430** may comprise a substrate **431**. The substrate **431** may be coupled to a lower surface of the base **410**. The substrate **431** may be coupled to the reinforcing member **420**. The substrate **431** may be coupled to the image sensor module **440**. The substrate **431** may comprise a hole **431-1**. The hole **431-1** may be coupled to the first protrusion **414** of the base **410**. The substrate **431** may comprise a protrusion **431-2**. The protrusion **431-2** may be formed to be protruded inward from a corner of the substrate **431**. A space in which the hole **431-1** is to be formed may be secured through the protrusion **431-2**. A hole **431-1** may be formed in the protrusion **431-2**.

(86) The terminal portion **430** may comprise a terminal **432**. The terminal **432** may be electrically connected to a terminal of the image sensor **444**. The terminal **432** may comprise a plurality of terminals. The terminal **432** may comprise a total of 24 terminals.

(87) The terminal **432** comprises a first coupling portion **432-1** disposed on the board **431**, a second coupling portion **432-2** coupled to the wire **510**, and a connection portion **432-3** connecting a first coupling portion **432-1** and the second coupling portion **432-2** to each other. A hole through which the wire **510** passes may be formed in the second coupling portion **432-2**. The second coupling portion **432-2** may be coupled to the wire **510** by soldering. The connection portion **432-3** may comprise a bent portion. The connection portion **432-3** may be bent a plurality of times. The connection portion **432-3** may have elasticity. The terminal **432** may have elasticity.

(88) The camera device **10A** may comprise an image sensor module **440**. The image sensor module **440** may be coupled to the base **410**. The image sensor module **440** may be fixed to the base **410**. The image sensor module **440** may move integrally with the base **410**. The image sensor module **440** may comprise a cover **441**, a filter **442**, a substrate **443**, an image sensor **444**, and a reinforcing plate **445**. However, any one or more among the cover **441**, the filter **442**, the substrate **443**, the image sensor **444**, and the reinforcing plate **445** of the image sensor module **440** may be omitted.

(89) The image sensor module **440** may comprise a cover **441**. The cover **441** may cover the filter **442** and the image sensor **444**. The cover **441** may comprise an upper plate portion and a side wall

portion. The cover **441** may comprise a hole **441a**. The hole **441a** may be a hollow hole. The hole **441a** may be an opening. The cover **441** may comprise a protrusion **441b**. The protrusion **441b** may be protruded from a lower surface of the cover **441**. The protrusion **441b** may be inserted into the second hole **443b** of the substrate **443** and the hole **445a** of the reinforcing plate **445**.

(90) The image sensor module **440** may comprise a filter **442**. The filter **442** may serve to block light of a specific frequency band from being incident on the image sensor **444** among the light passing through the lens module **210**. The filter **442** may be disposed parallel to the x-y plane. The filter **442** may be disposed between the lens module **210** and the image sensor **444**. The filter **442** may be disposed between the cover **441** and the substrate **443**. In a modified embodiment, the filter **442** may be disposed in the hole **441a** of the cover **441**. Filter **442** may comprise an infrared filter. The infrared filter may absorb or reflect infrared rays incident to the infrared filter.

(91) The image sensor module **440** may comprise a substrate **443**. The substrate **443** may be an 'image sensor substrate' on which the image sensor **444** is disposed. The substrate **443** may comprise a printed circuit board (PCB). The substrate **443** may comprise a circuit board. An image sensor **444** may be disposed on the substrate **443**. The substrate **443** may be coupled to the terminal portion **430**. The substrate **443** may comprise a first hole **443a** having a shape and size corresponding to that of the image sensor **444**. The image sensor **444** may be inserted into the first hole **443a** of the substrate **443**. The substrate **443** may comprise a second hole **443b**. The protrusion **441b** of the cover **441** may be inserted into the second hole **443b** of the substrate **443**. The substrate **443** may comprise a terminal **443c**. The terminal **443c** of the substrate **443** may be disposed at each of the four side end portions on a lower surface of the substrate **443**. The terminal **443c** of the substrate **443** may be connected to the terminal **432** of the terminal portion **430**. The substrate **443** may comprise a groove **443d**. The grooves **443d** of the substrate **443** may be formed in each of the four corners of the substrate **443**. The first protrusion **414** of the base **410** may be avoided by the groove **443d** of the substrate **443**.

(92) The image sensor module **440** may comprise an image sensor **444**. The image sensor **444** may be coupled to the base **410**. The image sensor **444** may move integrally with the base **410**. However, the image sensor **444** is not directly coupled to the base **410**, but the substrate **443** to which the image sensor **444** is coupled may be coupled to the base **410**. In a modified embodiment, the image sensor **444** may be coupled directly to the base **410**. The image sensor **444** may be disposed in alignment with the optical module. The image sensor **444** may have a configuration in which light passing through a lens and a filter **442** is incident, thereby forming an image. The image sensor **444** may be mounted on the substrate **443**. The image sensor **444** may be electrically connected to the substrate **443**. For example, the image sensor **444** may be coupled to the substrate **443** by a surface mounting technology (SMT). As another example, the image sensor **444** may be coupled to the substrate **443** by a flip chip technology. The image sensor **444** may be disposed so that the lens and the optical axis coincide. That is, the optical axis of the image sensor **444** and the optical axis of the lens may be aligned. The image sensor **444** may convert light irradiated to the effective image area of the image sensor **444** into an electrical signal. The image sensor **444** may be any one among a charge coupled device (CCD), a metal oxide semi-conductor (MOS), a CPD, and a CID.

(93) In the present embodiment, the image sensor **444** may be rotated about the x-axis, the y-axis, and the z-axis. The image sensor **444** may move about an x-axis, a y-axis, and a z-axis. The image sensor **444** may be tilted about an x-axis, a y-axis, and a z-axis.

(94) The image sensor module **440** may comprise a reinforcing plate **445**. The reinforcing plate **445** may be disposed on a lower surface of the image sensor **444** and the substrate **443**. The reinforcing plate **445** may be formed of SUS. The reinforcing plate **445** may reinforce the image sensor **444** and the substrate **443**. The reinforcing plate **445** may comprise a hole **445a**. The hole **445a** may be coupled to the protrusion **441b** of the cover **441**. The reinforcing plate **445** may comprise a groove **445b**. The grooves **445b** may be formed in each of the four corners of the reinforcing plate **445**.

The groove **445b** may be formed as a corner of the reinforcing plate **445** is inwardly recessed.

(95) The camera device **10A** may comprise a wire **510**. The wire **510** may connect the substrate **120** and the terminal portion **430**. The wire **510** may have elasticity. The wire **510** may be an elastic member. The wire **510** may be a wire spring. The wire **510** may be formed of metal. The wire **510** may be electrically connected to the image sensor **444**. The wire **510** may be used as a conductive line of the image sensor **444**. One end portion of the wire **510** may be coupled to the substrate **120** and the other end portion of the wire **510** may be coupled to the terminal **432**. The wire **510** may elastically support the movement of the base **410**.

(96) The wire **510** may comprise a plurality of wires. The plurality of wires may comprise a number of wires corresponding to the number of terminals of the image sensor **444**. The plurality of wires may comprise a total of 24 wires, 6 between adjacent corners among the 4 corners of the base.

(97) The camera device **10A** may comprise a sensor **520**. The sensor **520** may be disposed on an upper surface of the substrate **120**. The sensor **520** may comprise a Hall sensor (Hall IC). The sensor **520** may detect a magnetic force of the magnet **320**. The movement of the image sensor **444** may be detected in real time through the magnetic force of the magnet **320** detected by the sensor **520**. Through this, OIS feedback control may be possible.

(98) The sensor **520** may comprise a plurality of sensors. The sensor **520** may comprise three sensors. All of the x-axis direction movement, the y-axis direction movement, and the rotation about z-axis of the image sensor **444** may be detected through the three sensors. The sensor **520** may comprise first to third sensors. The first sensor may face the first magnet **321**, the second sensor may face the second magnet **322**, and the third sensor may face the third magnet **323**.

(99) The sensor **520** may comprise a first Hall sensor that detects the movement amount and/or displacement of the magnet **320** in the x-axis direction. The sensor **520** may comprise a second Hall sensor that detects the movement amount and/or displacement of the magnet **320** in the y-axis direction. The sensor **520** may comprise a third Hall sensor that detects the movement amount and/or displacement of the magnet **320** in the x-axis direction or the amount and/or displacement of the magnet **320** in the y-axis direction. A movement of the magnet **320** rotating about the z-axis may be detected through any two or more among the first Hall sensor, the second Hall sensor, and the third Hall sensor.

(100) The sensor **520** may comprise a first Hall sensor that detects the movement amount and/or displacement of the magnet **320** in the x-axis direction. The sensor **520** may comprise a second Hall sensor that detects the movement amount and/or displacement of the magnet **320** in the y-axis direction. The sensor **520** may comprise a third Hall sensor that detects the movement amount and/or displacement of the magnet **320** in the x-axis direction or the amount and/or displacement of the magnet **320** in the y-axis direction. A movement of the magnet **320** rotating about the z-axis may be detected through any two or more among the first Hall sensor, the second Hall sensor, and the third Hall sensor.

(101) The housing **600** may comprise a sidewall **610**. The sidewall **610** may comprise a plurality of sidewalls. The sidewall **610** may comprise four sidewalls. The housing **600** may comprise a lower portion **620**. The lower portion **620** may be extended inwardly from a lower end of the sidewall **610**. The lower portion **620** may comprise a hole. The lower surface of the housing **600** may be formed by a separate lower plate **630**. The lower plate **630** may be understood as one configuration or a separate configuration of the housing **600**. The lower plate **630** may comprise a groove **631** engaged with the protrusion **622** being protruded from a lower surface of the lower portion **620** of the housing **600**.

(102) The camera device **10A** may comprise a printed circuit board (PCB) **50**. The printed circuit board **50** may be electrically connected to the coil **310**. The printed circuit board **50** may comprise a terminal **50a** being coupled to the terminal **222** of the MEMS actuator **220**. The printed circuit board **50** may comprise a hole through which the lens module **210** passes.

(103) The camera device **10A** may comprise a connector **90**. The connector **90** may be electrically connected to the printed circuit board **50**. The connector **90** may comprise a port for electrically connecting to an external device.

(104) The camera device **10A** may comprise a motion sensor. The motion sensor may be mounted on the printed circuit board **50**. The motion sensor may be electrically connected to the control unit through a circuit pattern provided on the printed circuit board **50**. The motion sensor may output rotational angular velocity information due to the movement of the camera device **10A**. The motion sensor may comprise any one or more among a 2-axis gyro sensor, a 3-axis gyro sensor, and an angular velocity sensor.

(105) The camera device **10A** may comprise a control unit. The control unit may be disposed on the printed circuit board **50**. The control unit may be electrically connected to the coil **310**. The control unit may individually control the direction, intensity, and amplitude of the current supplied to the first to fourth coils **311**, **312**, **313**, and **314**. The control unit may control the current applied to the coil **310** and the current applied to the MEMS actuator **220** or the liquid lens **220a** to perform the autofocus function and/or the hand-shake correction function. Furthermore, the control unit may perform autofocus feedback control and/or hand-shake correction feedback control.

(106) The camera device **10A** according to the present embodiment may be the one for mobile camera application. That is, it can be distinguished from a camera device for digital camera application. When miniaturizing for mobile camera application, the driving force of the VCM is relatively lowered, so there is a problem in that the current consumption increases in order to implement the three operations (X-Shift, Y-Shift, and Z-Rotation (Roll)).

(107) The magnet **320** and the coil **310** are disposed at each corner of the base **410** being rotated 90 degrees from each other so that the magnet **320** and the coil **310** that are diagonally positioned can be assembled in the same direction. In this case, a Lorentz Force in the same direction may be generated when the image sensor **444** is driven for shift, and two pairs of rotational forces may be generated by a force in the opposite direction when the z-axis is driven for rotation.

(108) In the present embodiment, since the four coils located at the corners require current input independent of each other, it is possible to have a system in which the power terminal of the coil **310** is separated and controlled by four channels. That is, the present embodiment may comprise a diagonal arrangement of magnets in the same magnetic flux direction and an individual current input structure of four coils.

(109) The present embodiment may comprise two pairs of turning force generating structures (increasing rotational moment). With a structure that generates two pairs of rotational forces, it can generate a higher rotational moment than before, and the total current consumption can be reduced when operating in three modes: X-Shift, Y-Shift, and Z-Rotation (Roll).

(110) Simulation results for the camera device according to the present embodiment are as follows. When 'rotational moment=rotational force*distance between rotational forces=(electromagnetic force*input current)*distance between the centers of the magnet **320**', and when 50 mA is applied as an input current to the coil **310** of the camera device **10A** according to the present embodiment, it is confirmed that a rotational moment of $\{(0.094 \text{ mN/mA} \cdot 50 \text{ mA}) \cdot 12.14 \text{ mm}\} \cdot 2 = 114.1 \text{ mN} \cdot \text{mm}$ is generated.

(111) In the present embodiment, hand-shake correction for the image sensor **444** and hand-shake correction for a corresponding lens can be performed together. For example, when hand-shake correction is performed only with the MEMS actuator **220** or the liquid lens **220a**, positive (+) distortion may occur at the edge of the image obtained from the image sensor **444**. Meanwhile, when only the image sensor **444** is moved to perform hand-shake correction, negative (−) distortion may occur at the edge of the image obtained from the image sensor **444**. In the present embodiment, distortion occurring at the edge of the image can be minimized by performing hand-shake correction for the image sensor **444** and hand-shake correction in the MEMS actuator **220** or the liquid lens **220a** together. In the present embodiment, the hand-shake correction function may

be performed on the lens side through the MEMS actuator **220** or the liquid lens **220a** and the image sensor **444** can be moved in response thereto. Through this, it is possible to provide a level of hand-shake correction corresponding to the module movement method, which is a method in which the lens and the image sensor **444** move integrally. However, even in the present embodiment, the MEMS actuator **220** or the liquid lens **220a** may provide only the AF function and may perform the OIS function through the movement of the image sensor **444**.

(112) Hereinafter, an optical device according to the present embodiment will be described with reference to the drawings.

(113) FIG. **19** is a perspective view of an optical device according to the present embodiment, and FIG. **20** is a block diagram of an optical device illustrated in FIG. **19**.

(114) The optical device **10B** may be any one among a mobile phone, a mobile phone, a smart phone, a portable smart device, a digital camera, a laptop computer, a digital broadcasting terminal, a PDA (Personal Digital Assistants), a PMP (Portable Multimedia Player), and a navigation device. However, the type of the optical device **10B** is not limited thereto, and any device for photographing an image or a picture may be comprised in the optical device **10B**.

(115) The optical device **10B** may comprise a main body **850**. The main body **850** may have a bar shape. Alternatively, the main body **850** may have various structures such as a slide type, a folder type, a swing type, a swivel type, in which two or more sub-bodies are coupled to be relatively movable. The main body **850** may comprise a case (casing, housing, and cover) forming an outer appearance. For example, the main body **850** may comprise a front case **851** and a rear case **852**. In a space formed between the front case **851** and the rear case **852**, various electronic components of the optical device **10B** can be embedded. A display **751** may be disposed on one surface of the main body **850**. A camera **721** may be disposed on one or more surfaces of one surface of the main body **850** and the other surface disposed opposite to the one surface.

(116) The optical device **10B** may comprise a wireless communication unit **710**. The wireless communication unit **710** may comprise one or more modules that enable wireless communication between the optical device **10B** and the wireless communication system or between the optical device **10B** and the network in which the optical device **10B** is located. For example, the wireless communication unit **710** may comprise any one or more among a broadcast reception module **711**, a mobile communication module **712**, a wireless Internet module **713**, a short-range communication module **714**, and a location information module **715**.

(117) The optical device **10B** may comprise an A/V input unit **720**. The audio/video (A/V) input unit **720** is for inputting an audio signal or a video signal, and may comprise any one or more of a camera **721** and a microphone **722**. At this time, the camera **721** may comprise the camera device **10A** according to the present embodiment.

(118) The optical device **10B** may comprise a sensing unit **740**. The sensing unit **740** can generate a sensing signal for controlling the operation of the optical device **10B** by detecting the current state of the optical device **10B** such as the opening/closing state of the optical device **10B**, the position of the optical device **10B**, the presence or absence of user contact, the orientation of the optical device **10B**, acceleration/deceleration of the optical device **10B**, and the like. For example, when the optical device **10B** is in the form of a slide phone, it is possible to sense whether the slide phone is opened or closed. In addition, it may be responsible for a sensing function related to whether the power supply unit **790** is supplied with power, whether the interface unit **770** is coupled to an external device, and the like.

(119) The optical device **10B** may comprise an input/output unit **750**. The input/output unit **750** may be a configuration for generating an input or output related to visual, auditory, or tactile sense. The input/output unit **750** may generate input data for controlling the operation of the optical device **10B**, and may also output information processed by the optical device **10B**.

(120) The input/output unit **750** may comprise any one or more among a keypad unit **730**, a display **751**, a sound output module **752**, and a touch screen panel **753**. The keypad unit **730** may generate

input data in response to a keypad input. The display **751** may output an image photographed by the camera **721**. The display **751** may comprise a plurality of pixels whose color changes according to an electrical signal. For example, the display **751** may comprise at least one among a liquid crystal display, a thin film transistor-liquid crystal display, an organic light-emitting diode, a flexible display, or a three-dimensional display (3D display). The sound output module **752** may output audio data received from the wireless communication unit **710** in a call signal reception, a call mode, a recording mode, a voice recognition mode, or a broadcast reception mode, or may output audio data stored in the memory unit **760**. The touch screen panel **753** may convert a change in capacitance generated due to a user's touch on a specific area of the touch screen into an electrical input signal.

(121) The optical device **10B** may comprise a memory unit **760**. The memory unit **760** may store a program for processing and controlling the control unit **780**. In addition, the memory unit **760** may store input/output data, for example, any one or more among a phone book, a message, an audio, a still image, a photo, and a video. The memory unit **760** may store an image photographed by the camera **721**, for example, a photo or a video.

(122) The optical device **10B** may comprise an interface unit **770**. The interface unit **770** serves as a passage for connecting to an external device connected to the optical device **10B**. The interface unit **770** may receive data from an external device, receive power and transmit it to each component inside the optical device **10B**, or transmit data inside the optical device **10B** to an external device. The interface unit **770** may comprise any one or more among a wired/wireless headset port, an external charger port, a wired/wireless data port, a memory card port, a port for connecting a device equipped with an identification module, and an audio input/output (I/O) port, a video input/output (I/O) port, and an earphone port.

(123) The optical device **10B** may comprise a control unit **780**. The control unit **780** may control the overall operation of the optical device **10B**. The control unit **780** may perform related control and processing for voice call, data communication, video call, and the like. The control unit **780** may comprise a multimedia module **781** for playing multimedia. The multimedia module **781** may be provided inside the control unit **780** or may be provided separately from the control unit **780**. The control unit **780** may perform a pattern recognition process capable of recognizing a handwriting input or a drawing input performed on the touch screen as characters and images, respectively.

(124) The optical device **10B** may comprise a power supply unit **790**. The power supply unit **790** may receive external power or internal power under the control of the control unit **780** to supply power required for operation of each component.

(125) The embodiments of the present invention have been described above with reference to the accompanying drawings, but a person skilled in the art to which the present invention belongs may understand that the present invention can be implemented in other specific forms without changing the technical spirit or essential features. Therefore, it should be understood that the embodiments described above are illustrative and non-limiting in all respects.

Claims

1. A camera device, comprising: a first substrate; a second substrate spaced apart from the first substrate; a wire configured to support the second substrate to move with respect to the first substrate; an image sensor and an infrared filter configured to move together with the second substrate; and a coil and a magnet configured to move the image sensor, wherein the magnet comprises first to third magnets spaced apart from each other, wherein the coil comprises a first coil facing the first magnet, a second coil facing the second magnet, and a third coil facing the third magnet, wherein the wire is disposed parallel to an optical axis of the image sensor, and wherein the wire is overlapped with the infrared filter in a direction perpendicular to the optical axis.

2. The camera device of claim 1, wherein the image sensor is configured to rotate about the optical axis by an interaction between the first to third magnets and the first to third coils.
3. The camera device of claim 1, comprising an elastic member comprising a coupling portion coupled with the wire, and a connection portion connected with the coupling portion, and wherein the connection portion comprises a bent portion and has an elasticity.
4. The camera device of claim 1, comprising: a holder assembly comprising the first substrate; and a base assembly comprising the second substrate, wherein an upper end portion of the wire is coupled with the holder assembly, and wherein a lower end portion of the wire is coupled with the base assembly.
5. The camera device of claim 1, comprising a voice coil motor (VCM) actuator to perform an auto focus function.
6. The camera device of claim 1, comprising a reinforcing plate configured to move together with the image sensor.
7. The camera device of claim 6, wherein the image sensor is directly coupled with an upper surface of the reinforcing plate.
8. The camera device of claim 7, wherein the second substrate comprises a hole, and wherein at least a portion of the image sensor is disposed in the hole of the second substrate.
9. The camera device of claim 8, comprising: a lens; and a cover disposed on the second substrate wherein the filter is disposed on an optical path formed between the lens and the image sensor.
10. The camera device of claim 1, wherein an upper surface of the second substrate comprises first to fourth corner areas, and wherein the first coil is disposed at the first corner area.
11. The camera device of claim 1, wherein the first to third coils are electrically separated from each other.
12. The camera device of claim 1, wherein the wire is electrically connected with the image sensor.
13. The camera device of claim 1, wherein the third coil and the third magnet are configured to move the image sensor in a first direction perpendicular to the optical axis, and wherein the second coil and the second magnet are configured to move the image sensor in a second direction perpendicular to each of the optical axis and the first direction.
14. The camera device of claim 13, wherein the third coil is not overlapped with the first coil in the first direction.
15. The camera device of claim 13, wherein the second coil is overlapped with the first coil in the second direction.
16. An optical device, comprising: a main body; the camera device of claim 1 disposed on the main body; and a display disposed on the main body and configured to output an image photographed by the camera device.
17. The camera device of claim 1, wherein the second substrate faces the first substrate in a direction of the optical axis, wherein the first to third magnets are disposed on the second substrate, wherein the first to third coils are disposed on the first substrate, wherein a length of the first coil measured in a first direction perpendicular to the optical axis is shorter than a width of the first coil measured in a second direction perpendicular to each of the first direction and the optical axis, and wherein a length of the second coil measured in the first direction is longer than a width of the second coil measured in the second direction.
18. A camera device, comprising: a first substrate; a second substrate configured to move with respect to the first substrate; an image sensor and an infrared filter configured to move together with the second substrate; a wire disposed parallel to an optical axis of the image sensor; and a coil and a magnet configured to move the image sensor, wherein the magnet comprises first to third magnets spaced apart from each other, wherein the coil comprises a first coil facing the first magnet in a direction parallel to the optical axis, a second coil facing the second magnet in a direction parallel to the optical axis, and a third coil facing the third magnet in a direction parallel to the optical axis, wherein the image sensor is configured to rotate about the optical axis by an

interaction between the first to third magnets and the first to third coils, and wherein the wire is overlapped with the infrared filter in a direction perpendicular to the optical axis.

19. The camera device of claim 18, wherein the third coil and the third magnet are configured to move the image sensor in a first direction perpendicular to the optical axis, and wherein the second coil and the second magnet are configured to move the image sensor in a second direction perpendicular to each of the optical axis and the first direction.

20. The camera device of claim 19, wherein the third coil is not overlapped with the first coil in the first direction, and wherein the second coil is overlapped with the first coil in the second direction.
